

4 . 3 D A N D V I D E O

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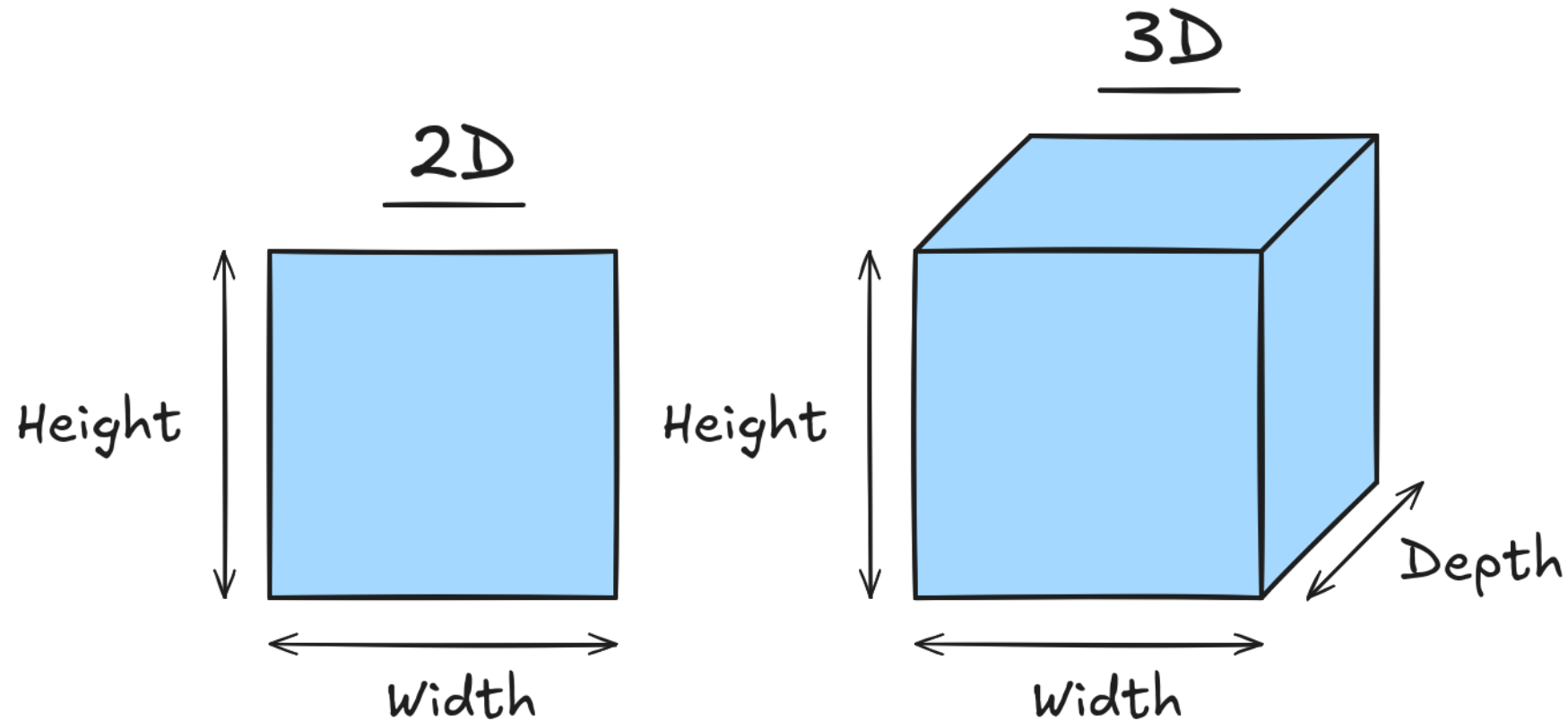
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1 . W H Y 3 D

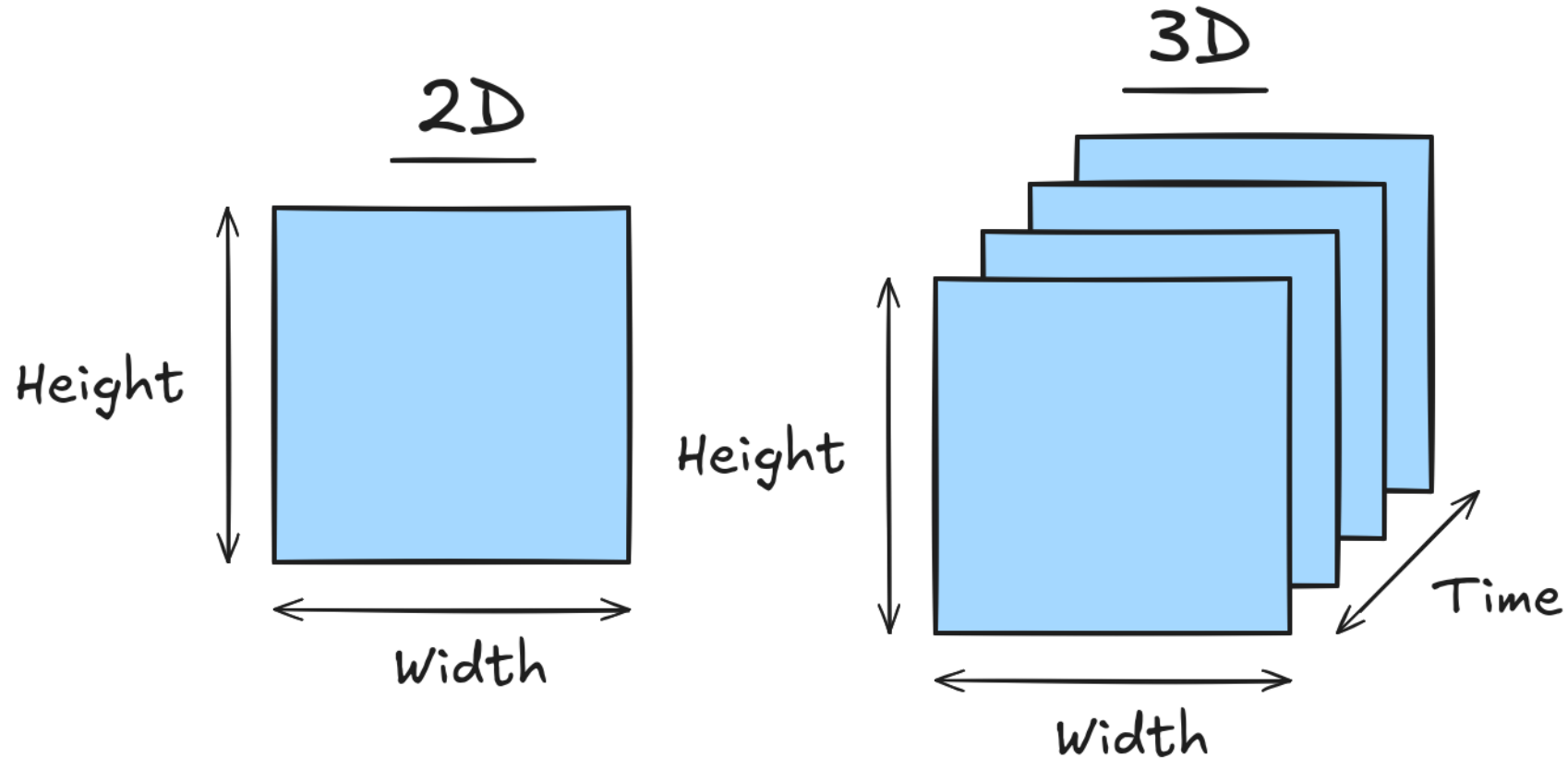
NEW SPATIAL DIMENSION



Taken the channel dimension into account it would be 3D and 4D.



NEW TEMPORAL DIMENSION

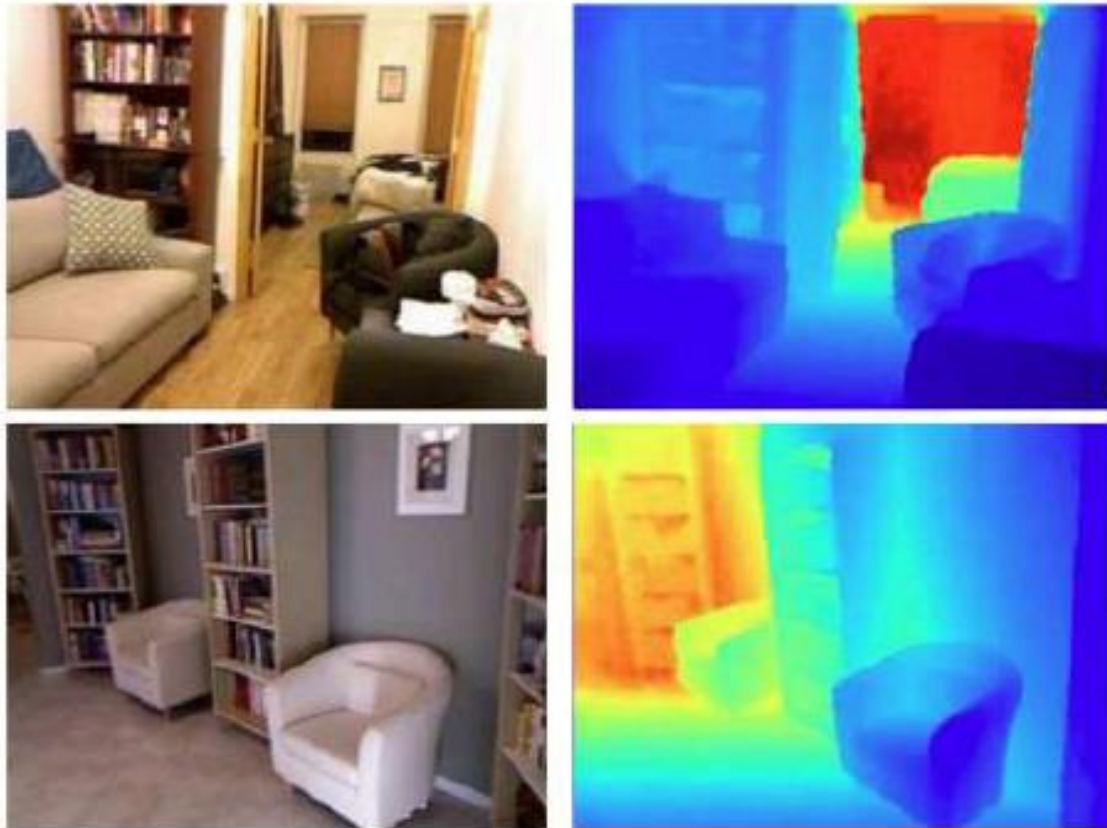


Taken the channel dimension into account it would be 3D and 4D.



2 . 3 D R E P R E S E N T A T I O N S

DEPTH MAP

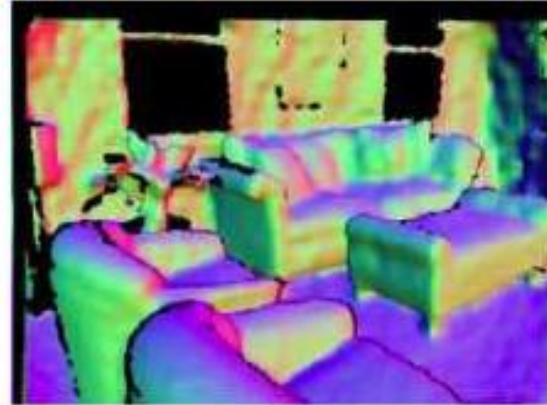
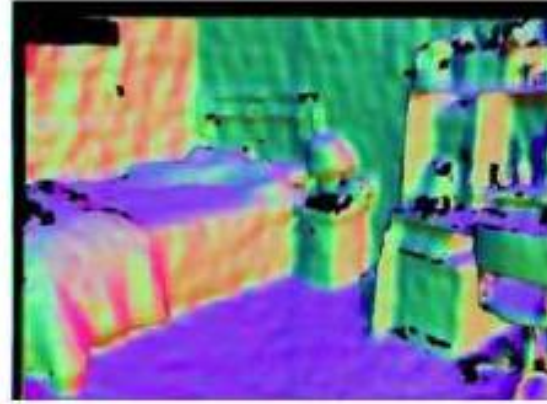


RGB Image: $3 \times H \times W$ Depth Map: $H \times W$

- Each pixel stores information about the distance between the camera and objects in the scene in the real world.
- Small values represent points close to the camera, and large values indicate points farther away.



SURFACE NORMALS



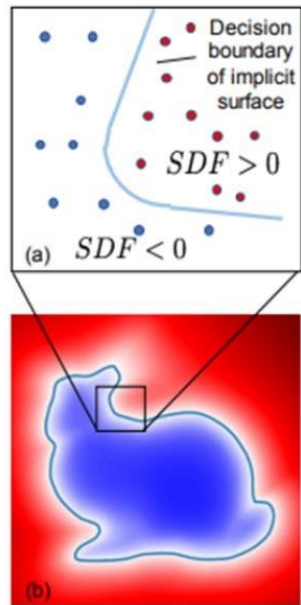
RGB Image: $3 \times H \times W$

Normals: $3 \times H \times W$

- Each pixel stores the orientation of the surface at that point.
- Each pixel is encoded as a 3D vector (x, y, z) .



IMPLICIT FUNCTIONS



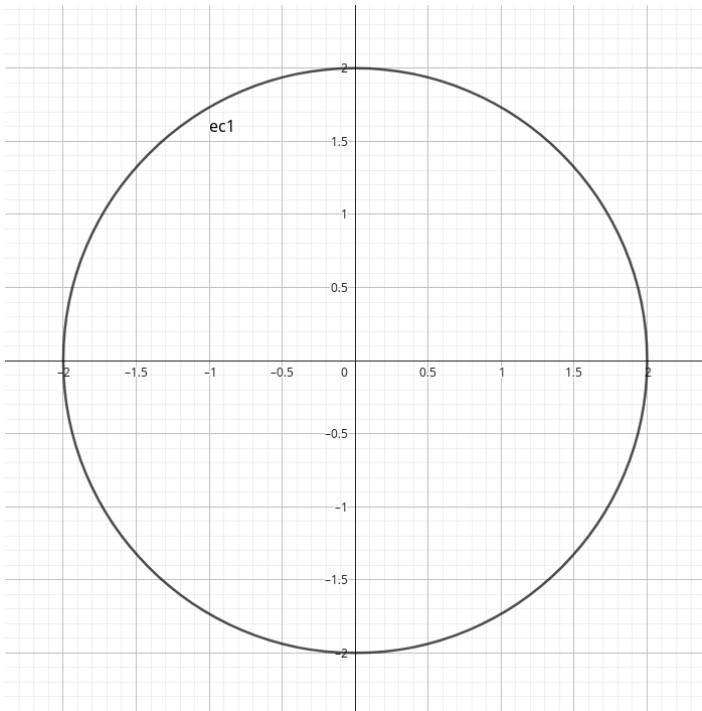
(c)



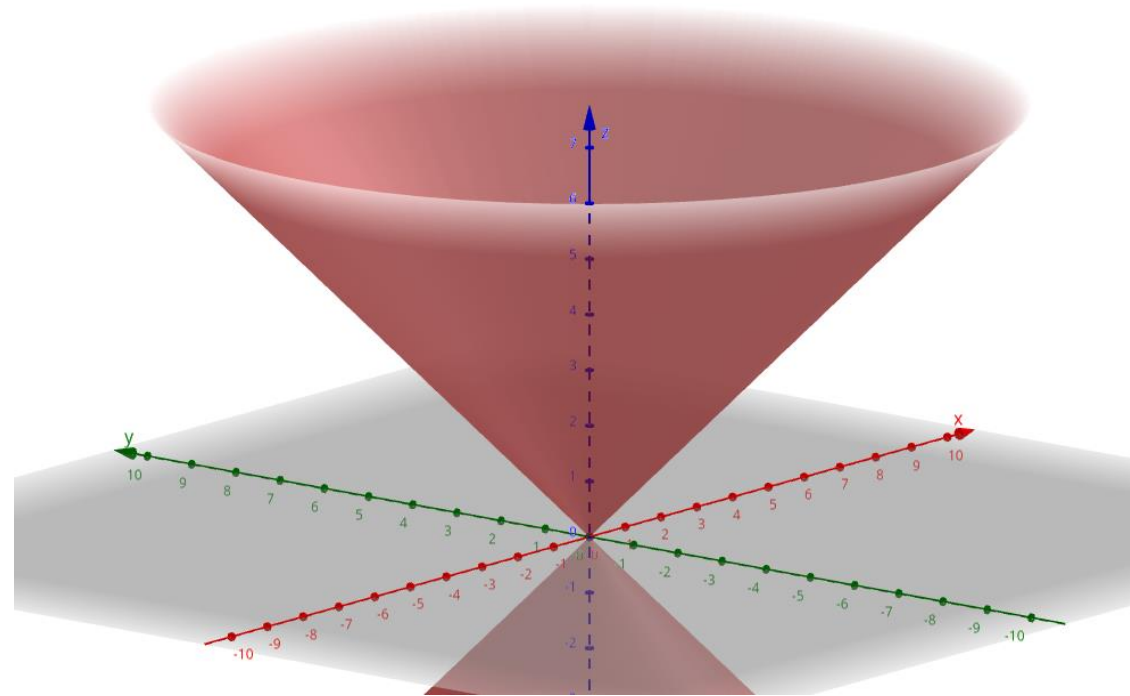
- A mathematical function $f(x, y, z)$ that defines a surface or volume continuously in space.
- Surface: $f(x, y, z) = 0$
- We can interpret $f(x, y, z)$ as the distance (not necessarily Euclidean) to the surface.



IMPLICIT FUNCTIONS



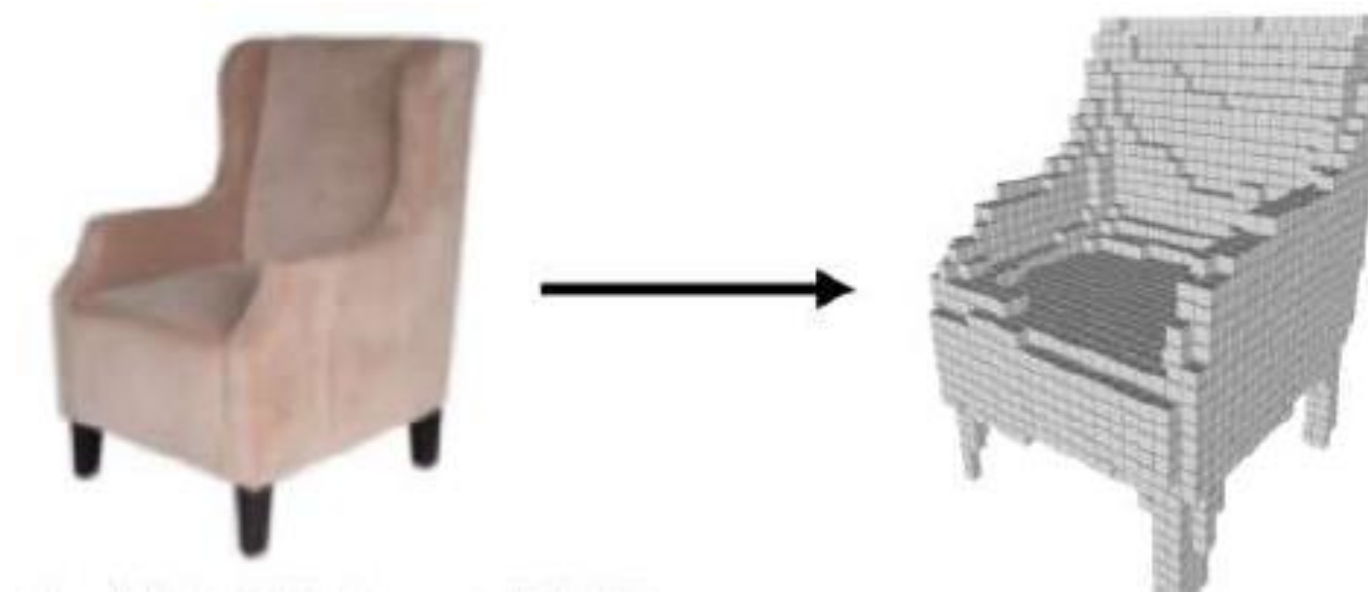
$$x^2 + y^2 - 4 = 0$$



$$x^2 + y^2 - z^2 = 0$$



VOXELS



- A pixel represent a point (small square) in 2D space; a voxel represents a point (small cube) in 3D space.
- Voxel-based structure divides a three-dimensional volume into a regular grid of cubes.
- Each voxel can store information. If they store n features, then the grid is $V \times V \times V \times n$.
- Spatial and memory inefficient.



POINT CLOUD

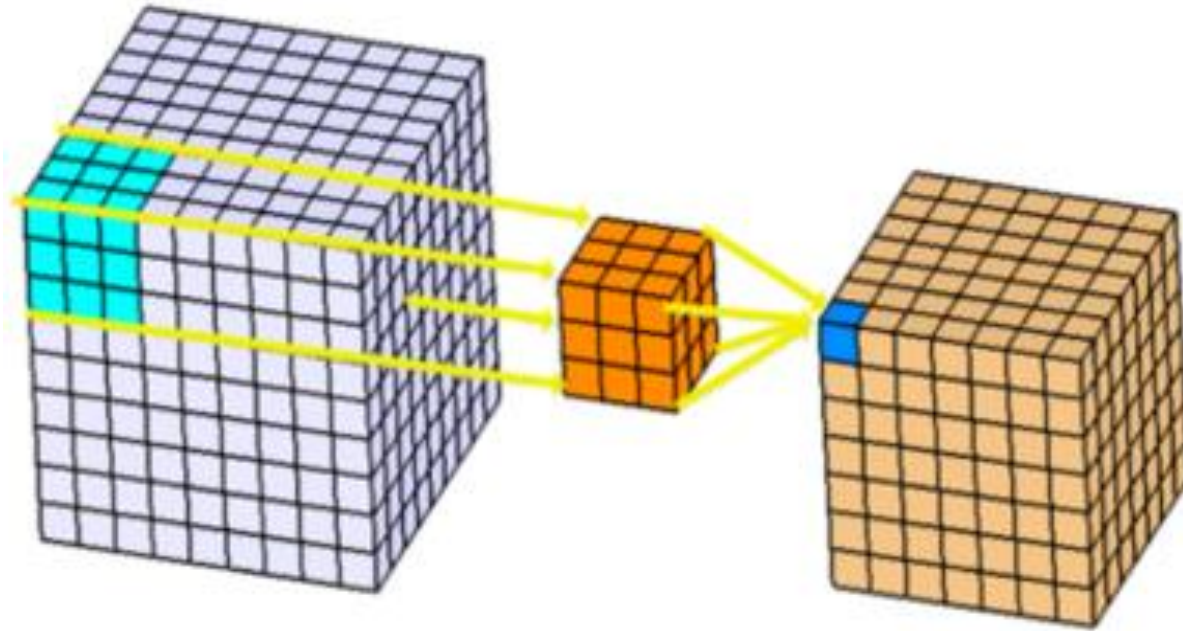


- Representation of a 3D shape or scene as a set of points in space.
- Like voxels, but sparse (therefore more efficient) and more flexible!
- Difficult to understand for networks.



3 . 3 D C O N V O L U T I O N S

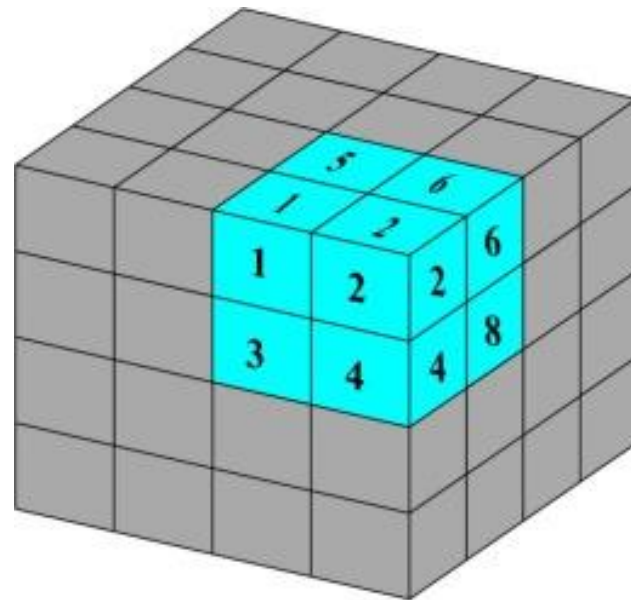
MODEL



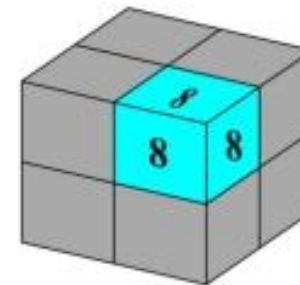
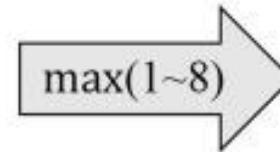
- Like 2D convolutions, but they also consider depth or temporal axis.
- Have high computational and memory cost.



MAX/AVERAGE POOLING (3D)



(a) Before max pooling



(b) After max pooling



4 . P R E D I C T I N G R E P R E S E N T A T I O N S

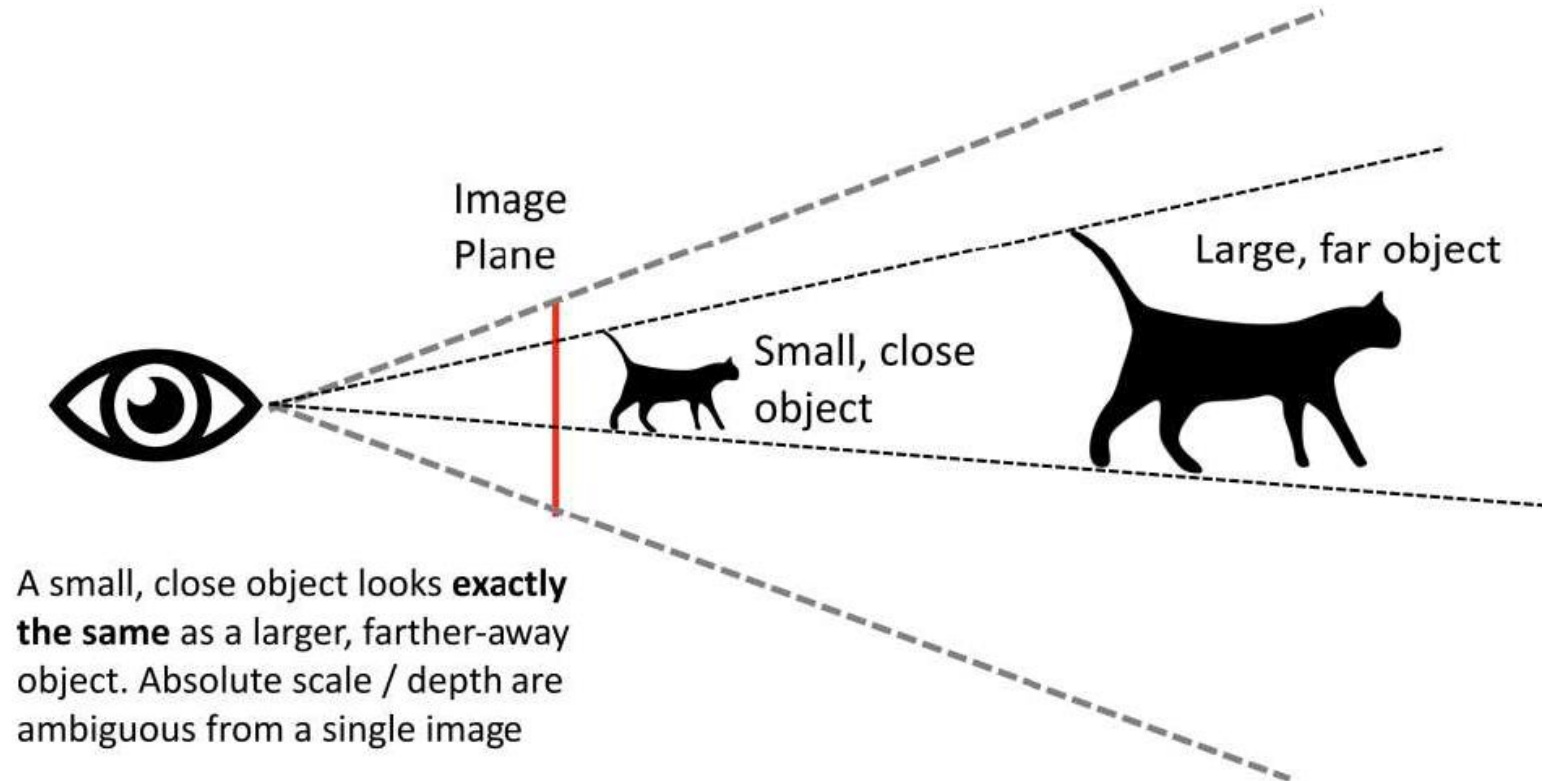
The diagram illustrates a residual block. An input layer (represented by a stack of three vertical rectangles) splits into two parallel paths. The main path consists of two sequential convolutional layers, each represented by a stack of three vertical rectangles. The residual path consists of a single convolutional layer, also represented by a stack of three vertical rectangles. Arrows indicate the flow of data from the input to both paths. The outputs of the two paths are then combined at the output layer (represented by a stack of three vertical rectangles) via a residual connection, which is shown as a curved arrow bypassing the main path's layers.



Per-Pixel Loss (L2 Distance)



PREDICTING DEPTH MAPS



PREDICTING DEPTH MAPS

$$\mathcal{L}(y, \hat{y}) = \underbrace{\frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W (\log(y_{ij}) - \log(\hat{y}_{ij}))^2}_{\text{prediction error}} - \underbrace{\frac{\lambda}{(HW)^2} \left(\sum_{i=1}^H \sum_{j=1}^W (\log(y_{ij}) - \log(\hat{y}_{ij})) \right)^2}_{\text{compensation for scale in prediction error}}$$

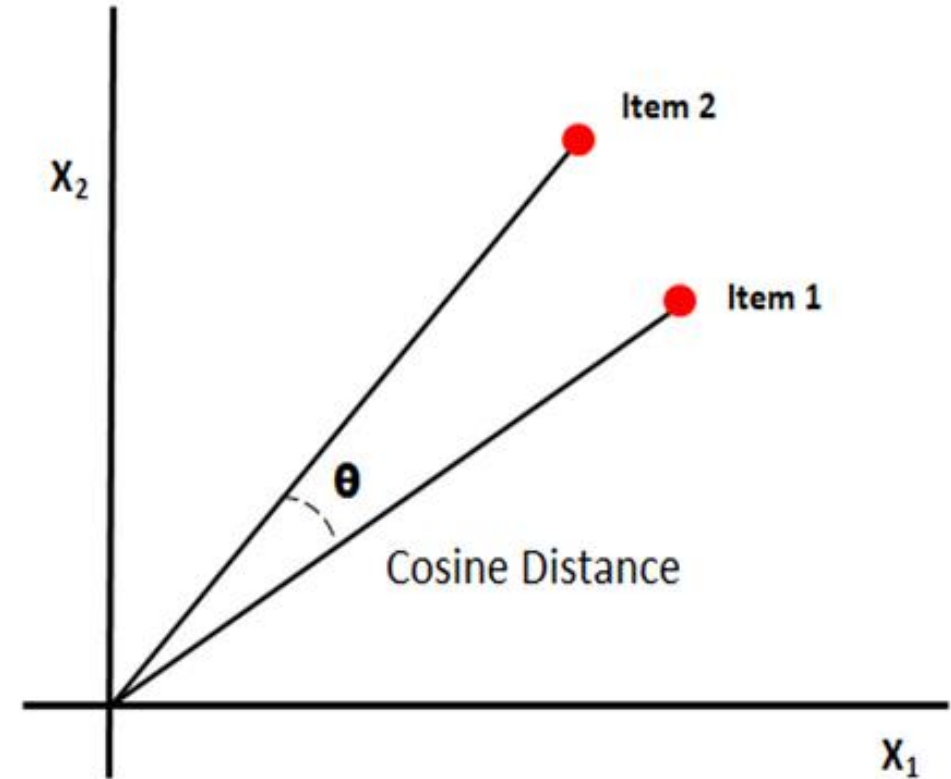
- λ {
- 0: Totally depends on the scale.
 - Between 0 and 1: Compensation term.
 - 1: Scale-invariant. $\forall c > 0 \implies \mathcal{L}(y, \hat{y}) = \mathcal{L}(y, c\hat{y})$

PREDICTING SURFACE NORMALS

Equal to predicting depth maps, but:

- FCN outputs 3 channels.
- We use cosine distance as loss function.

$$\mathcal{L}(y, \hat{y}) = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W \left(1 - \frac{\vec{y}_{ij} \cdot \vec{\hat{y}}_{ij}}{\|\vec{y}_{ij}\|_2 \|\vec{\hat{y}}_{ij}\|_2} \right)$$



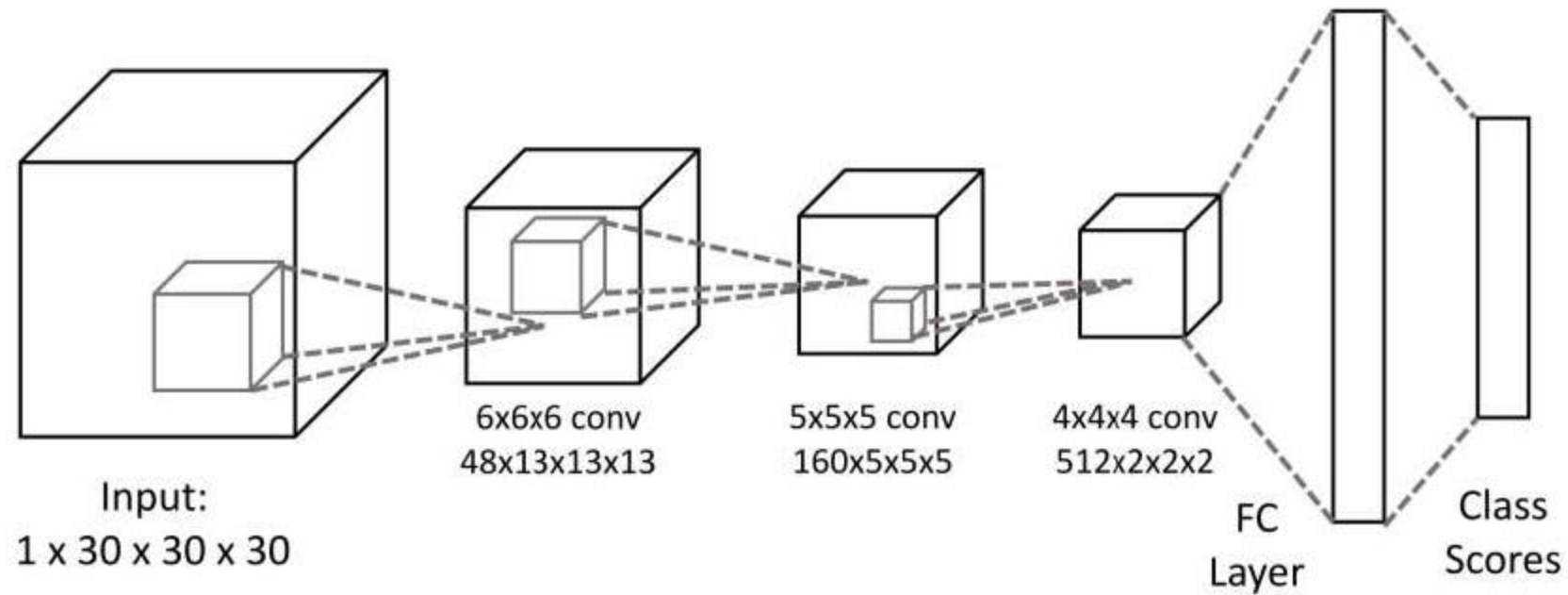
SIMULATING IMPLICIT FUNCTIONS

1. Sample random points in the space.
2. Compute the desired labels for each point (for example, Euclidean distance).
3. Train any model with the samples (points sampled) and labels.

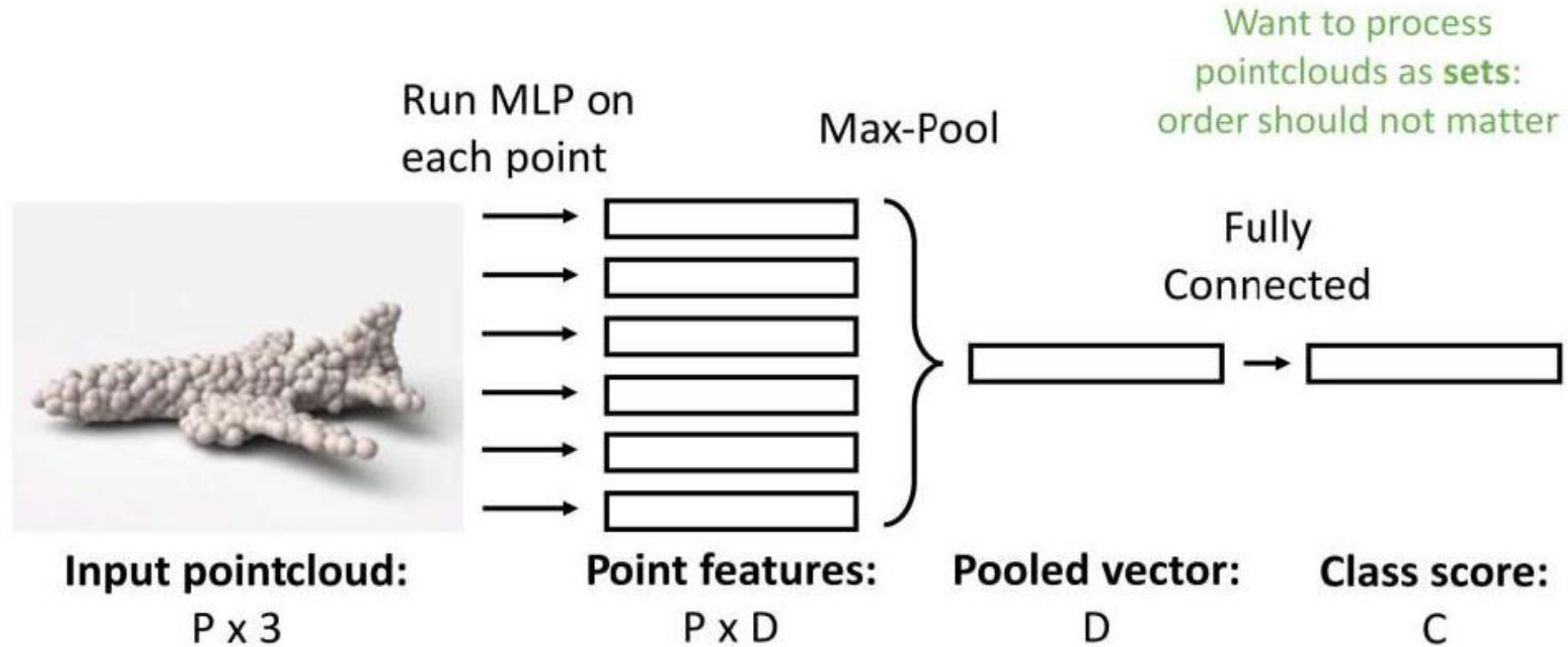
The output of the ML model is just one scalar, because we are simulating the implicit function.



PREDICTING VOXELS CLASSES



PREDICTING POINT CLOUDS



5 . P R E D I C T I N G S H A P E S

INTERSECTION OVER UNION (IOU)

$$d_{\text{IoU}}(P, Q) = \frac{|P \cap Q|}{|P \cup Q|}$$

- It can only be applied to voxels (because they are finite).
- Not good for thin structures.



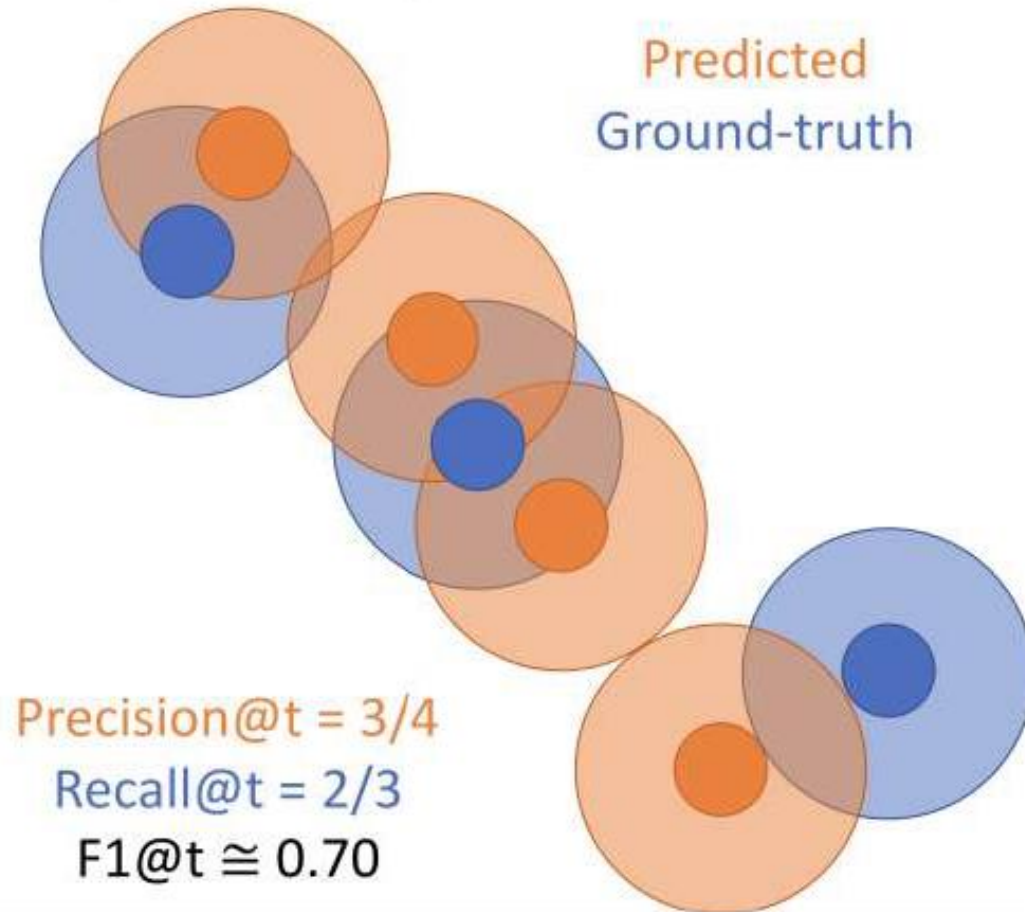
CHAMFER DISTANCE

$$d_C(P, Q) = \frac{1}{|P|} \sum_{p \in P} \underbrace{\min_{q \in Q} \{\|p - q\|_2^2\}}_{\text{closest point in } Q} + \frac{1}{|Q|} \sum_{q \in Q} \underbrace{\min_{p \in P} \{\|q - p\|_2^2\}}_{\text{closest point in } P}$$

- Symmetric, differentiable and it can be applied to any representation!
- It can be sensitive to outliers.



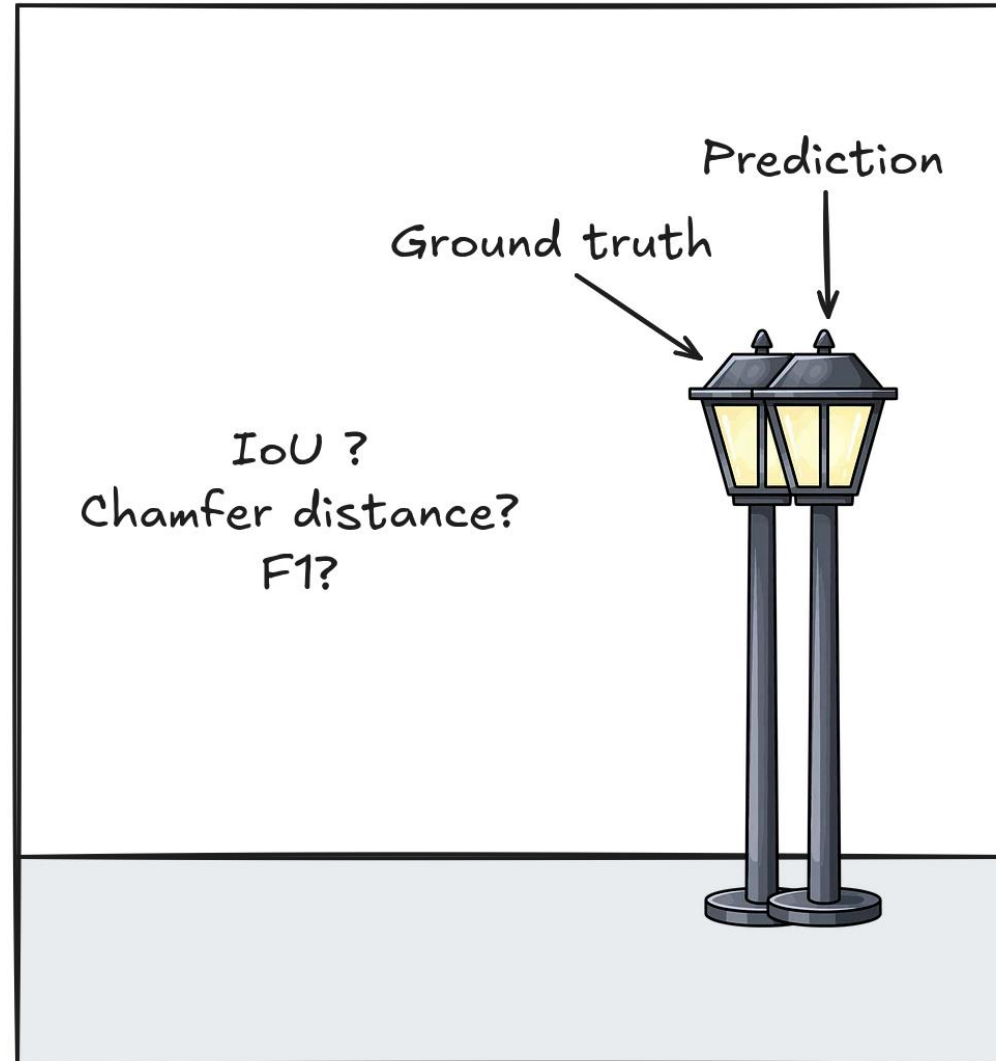
F1 SCORE



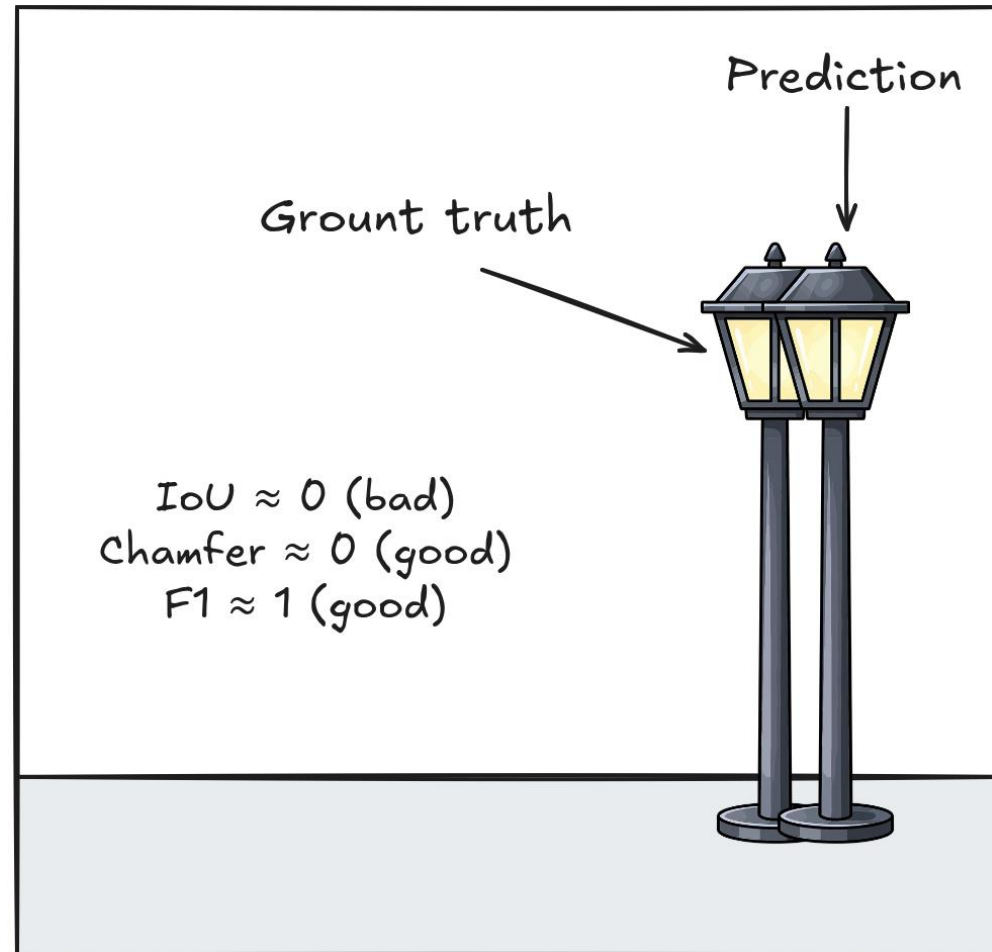
- It can be sensitive to the threshold.
- It is believed to be the best.
- It can be applied to any representation.



COMPARISON



COMPARATION



DATASETS

- Synthetic data
- Polygonal meshes

ShapeNet
Synthetic

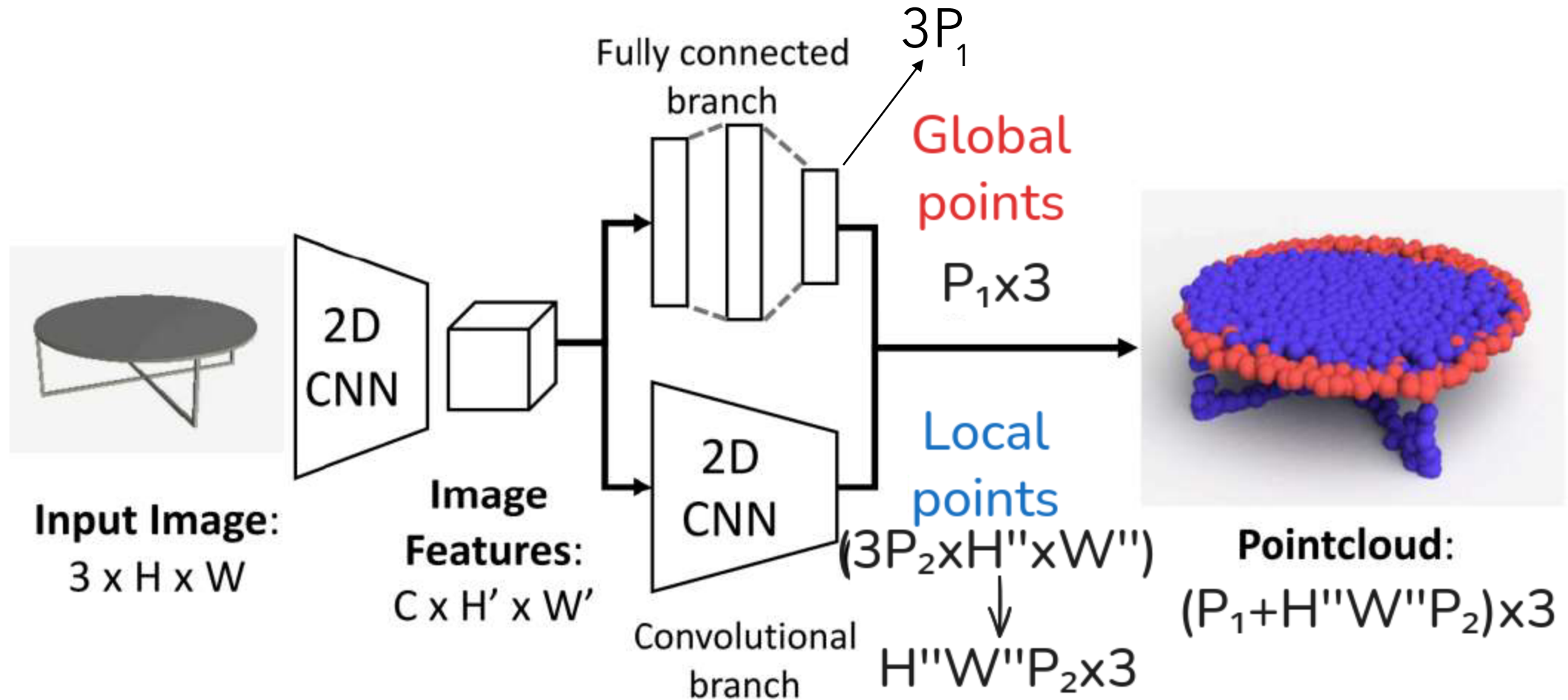


- Real world examples
- Images and its 3D models

Pix3D



GENERATING POINT CLOUDS



6 . V I D E O

INTRODUCTION TO VIDEO



Frame 1

...



Frame T//2

...



Frame T

$T \times C \times H \times W$

- Now we have information about motion and the evolution of objects over time!
- Videos are much heavier than images, so we may reduce resolution and fps.

ACTION RECOGNITION



Images: Recognize **objects**



Dog
Cat
Fish
Truck



Videos: Recognize **actions**

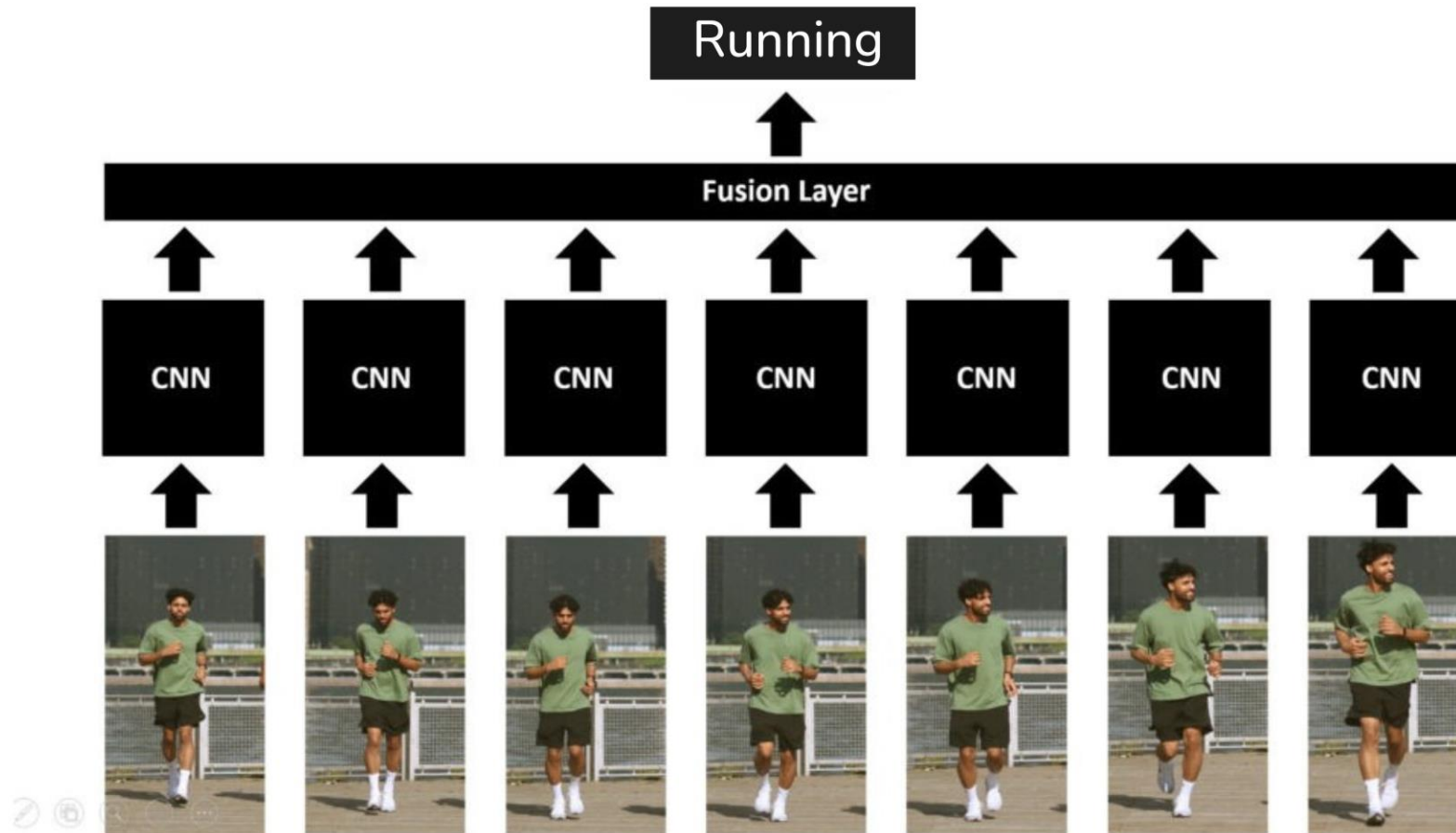


Swimming
Running
Jumping
Eating
Standing

- This task requires understanding also the **temporal evolution** of movements.
- **Intra-class** difficulties: The same action can be executed in very different ways depending on the person, speed, camera angle, or lighting condition.
- **Inter-class** difficulties: Different actions can be very similar (e.g., eat and drink).

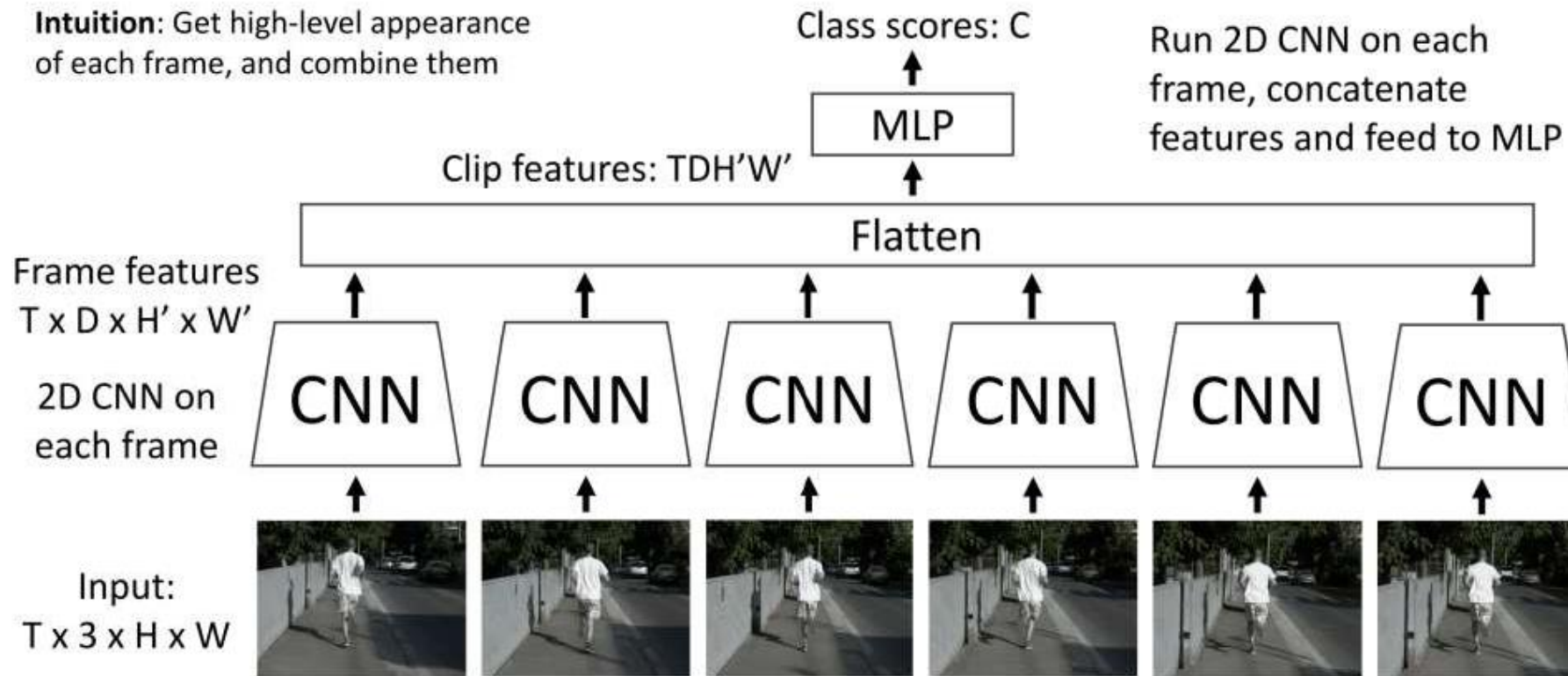


SINGLE-FRAME CNN



Simple, but works good and it is a decent baseline!

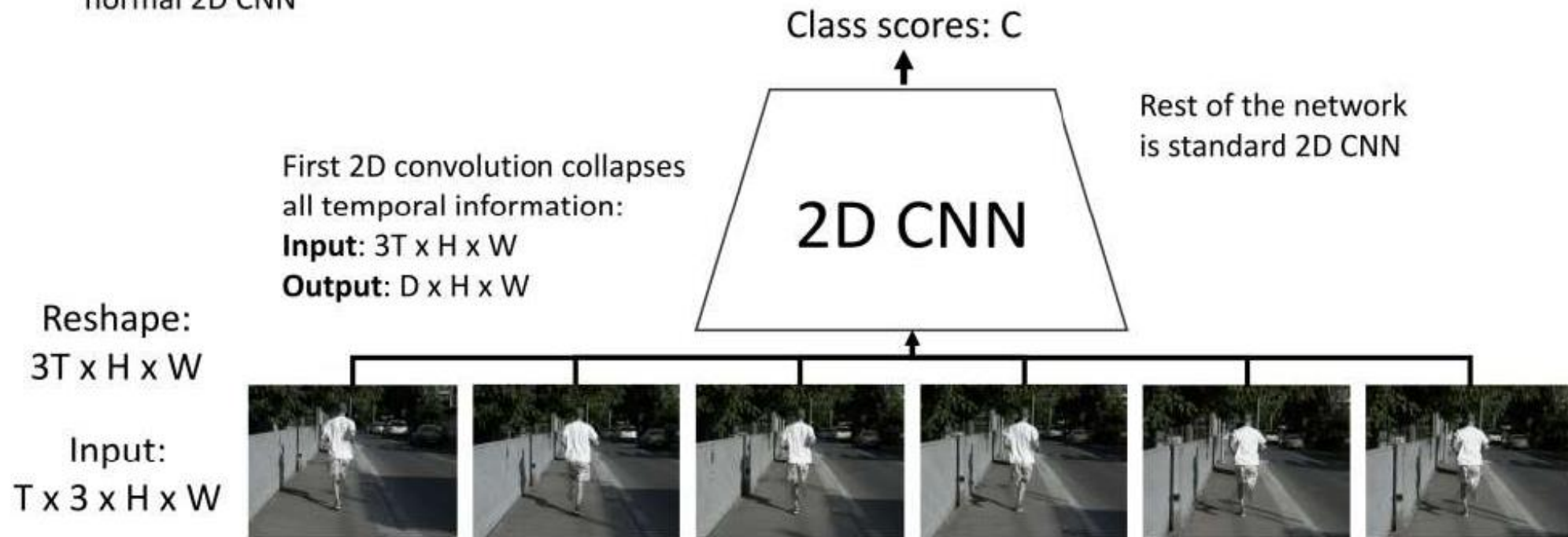
LATE FUSION



- We fuse images after CNN.
- We can take the mean over T , H' and W' dimensions to avoid the input being too large.
- Possible problem: It is hard to compare low-level motion between frames.

EARLY FUSION

Intuition: Compare frames with very first conv layer, after that normal 2D CNN



- We fuse images before CNN.
- Possible problem: may be too many input channels ?

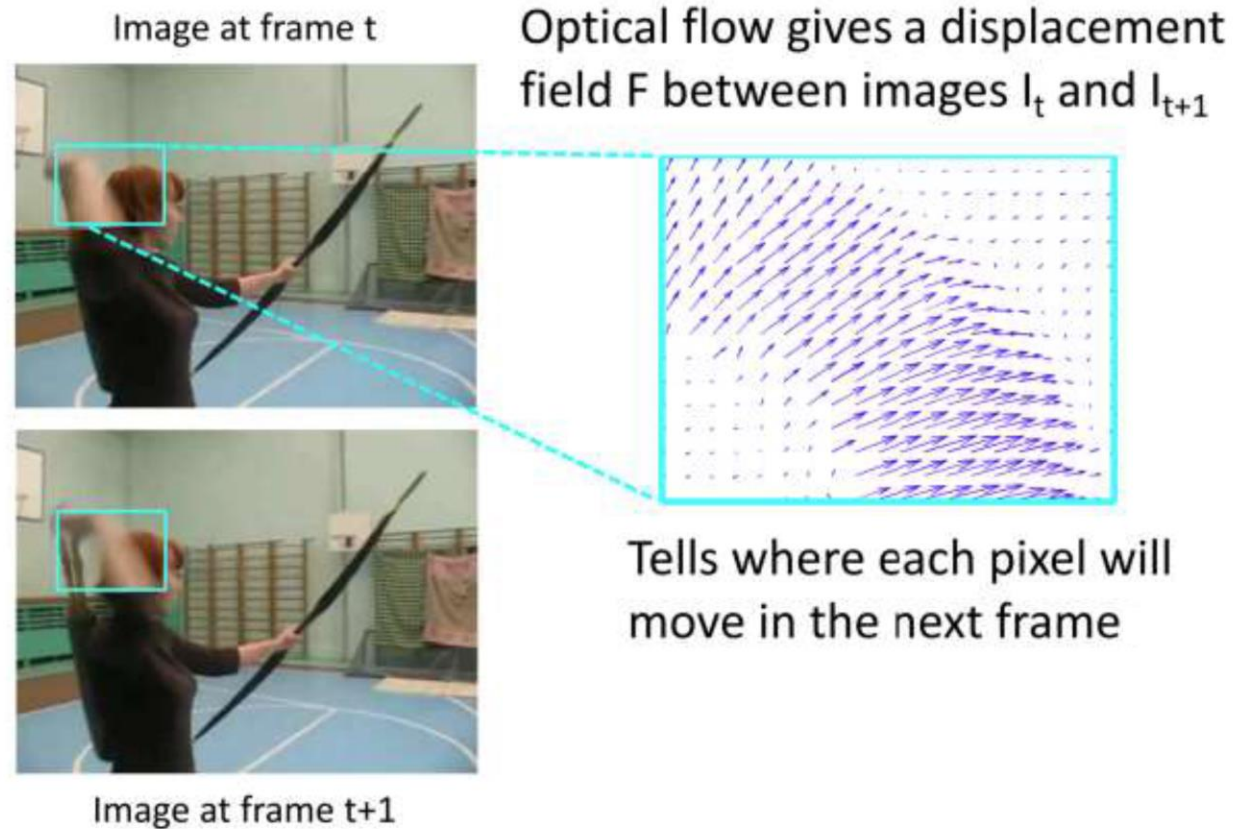
3D CNN

Layer	Size
Input	3 x 16 x 112 x 112
Conv1 (3x3x3)	64 x 16 x 112 x 112
Pool1 (1x2x2)	64 x 16 x 56 x 56
Conv2 (3x3x3)	128 x 16 x 56 x 56
Pool2 (2x2x2)	128 x 8 x 28 x 28
Conv3a (3x3x3)	256 x 8 x 28 x 28
Conv3b (3x3x3)	256 x 8 x 28 x 28
Pool3 (2x2x2)	256 x 4 x 14 x 14
Conv4a (3x3x3)	512 x 4 x 14 x 14
Conv4b (3x3x3)	512 x 4 x 14 x 14
Pool4 (2x2x2)	512 x 2 x 7 x 7
Conv5a (3x3x3)	512 x 2 x 7 x 7
Conv5b (3x3x3)	512 x 2 x 7 x 7
Pool5	512 x 1 x 3 x 3
FC6	4096
FC7	4096
FC8	C

The best, but very expensive computationally!



TWO-STREAM FLOW NETWORKS

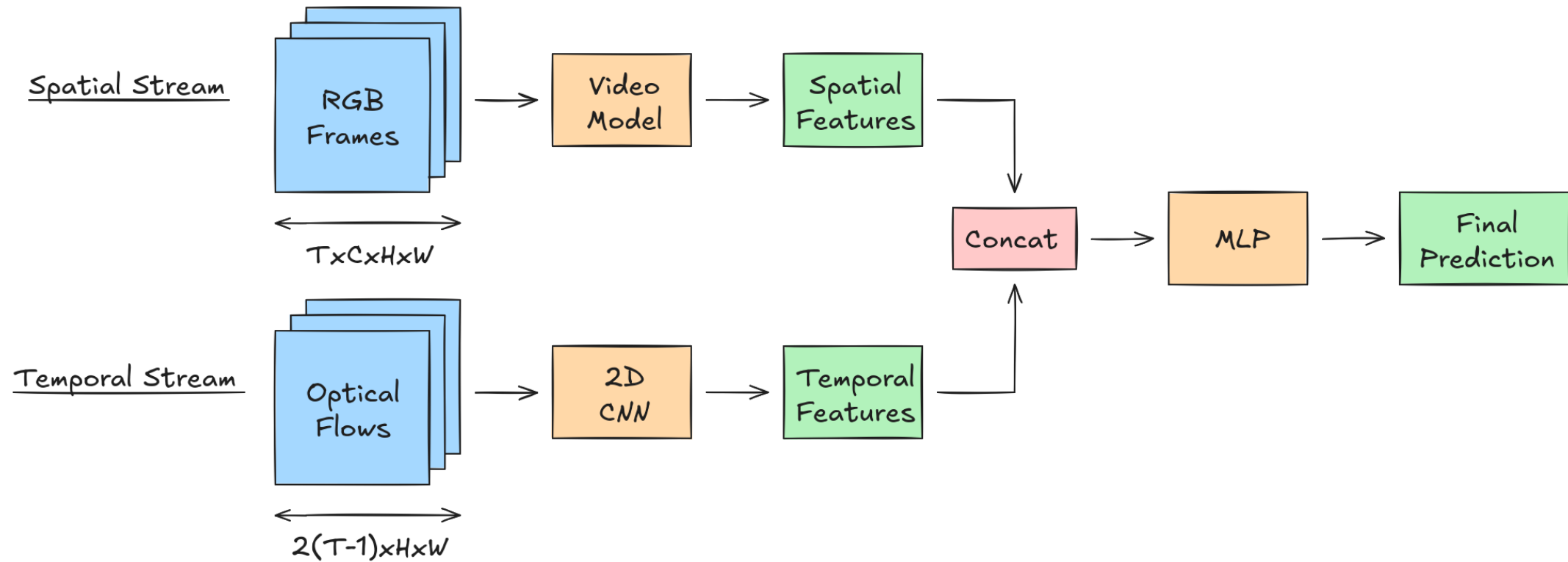


$$F_t(x, y) = (dx_t, dy_t)$$

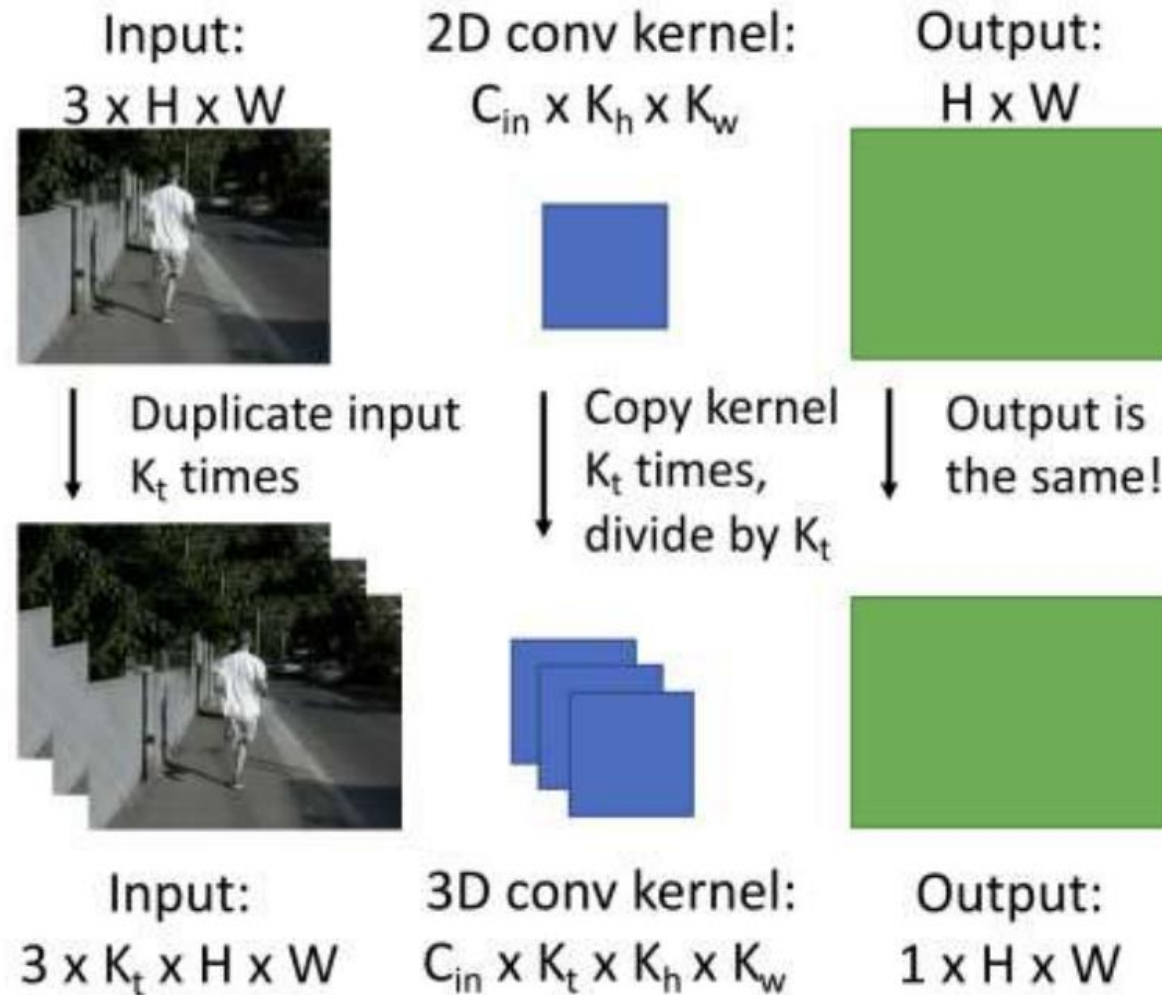
$$I_{t+1}(x, y) = I_t(x, y) + F_t(x, y) = (x_t + dx_t, y_t + dy_t)$$



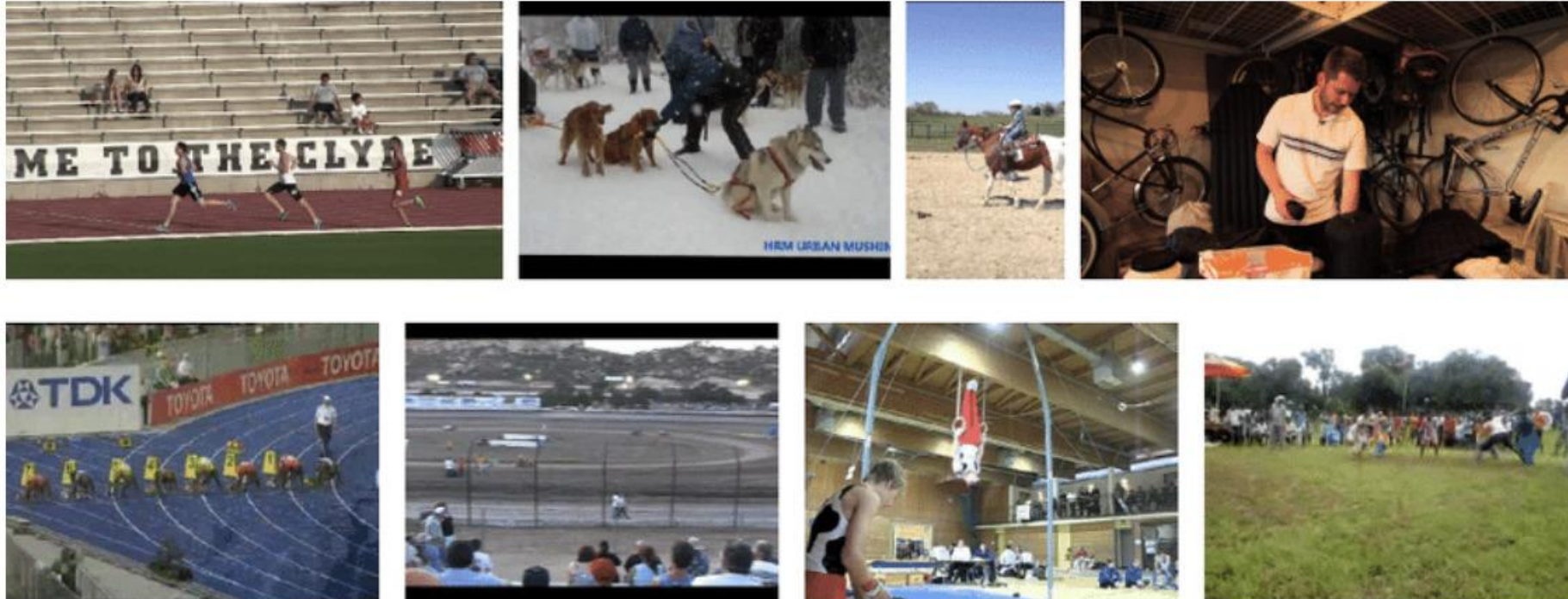
TWO-STREAM FLOW NETWORKS



INFLATED CNNs



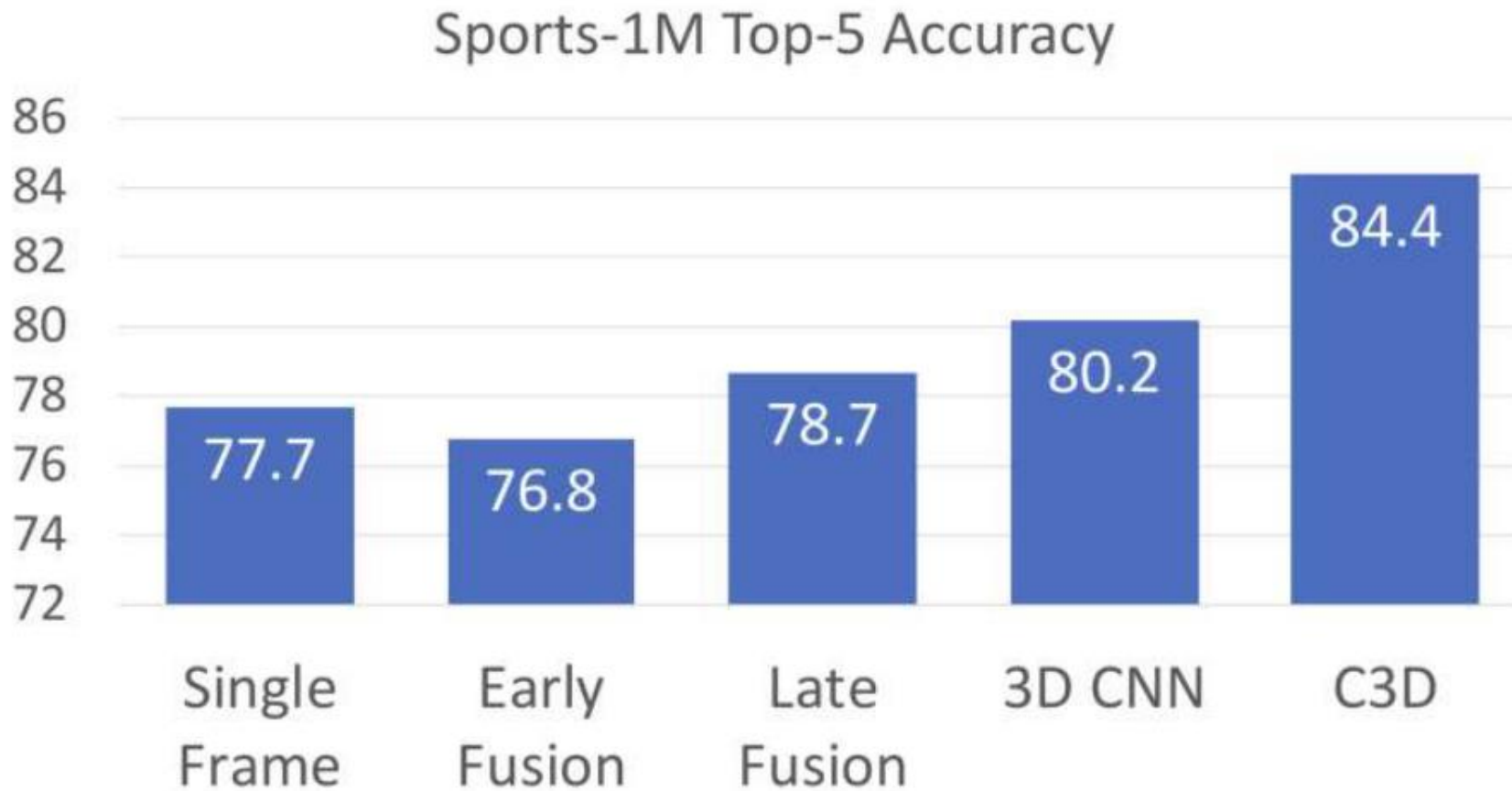
SPORTS-1M DATASET



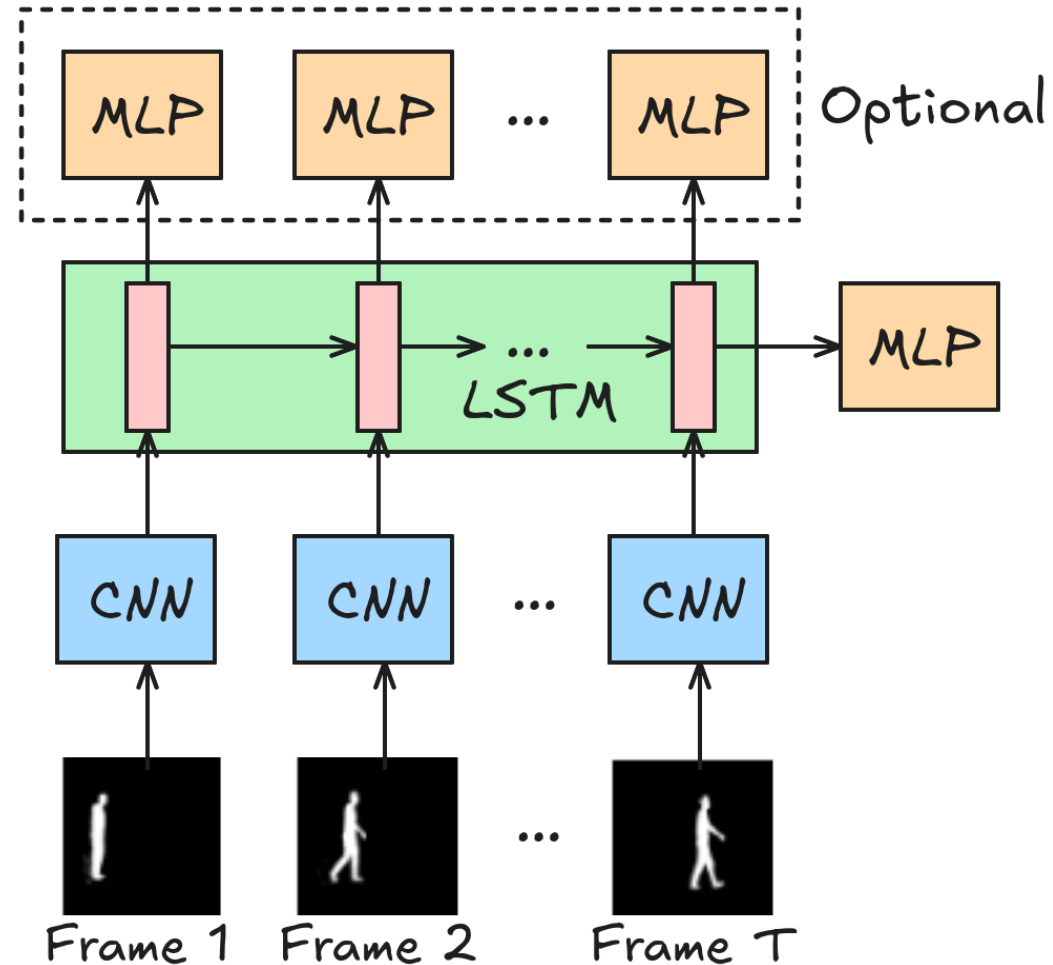
~ 1 M videos
~ 5 minutes per video
~ 500 classes



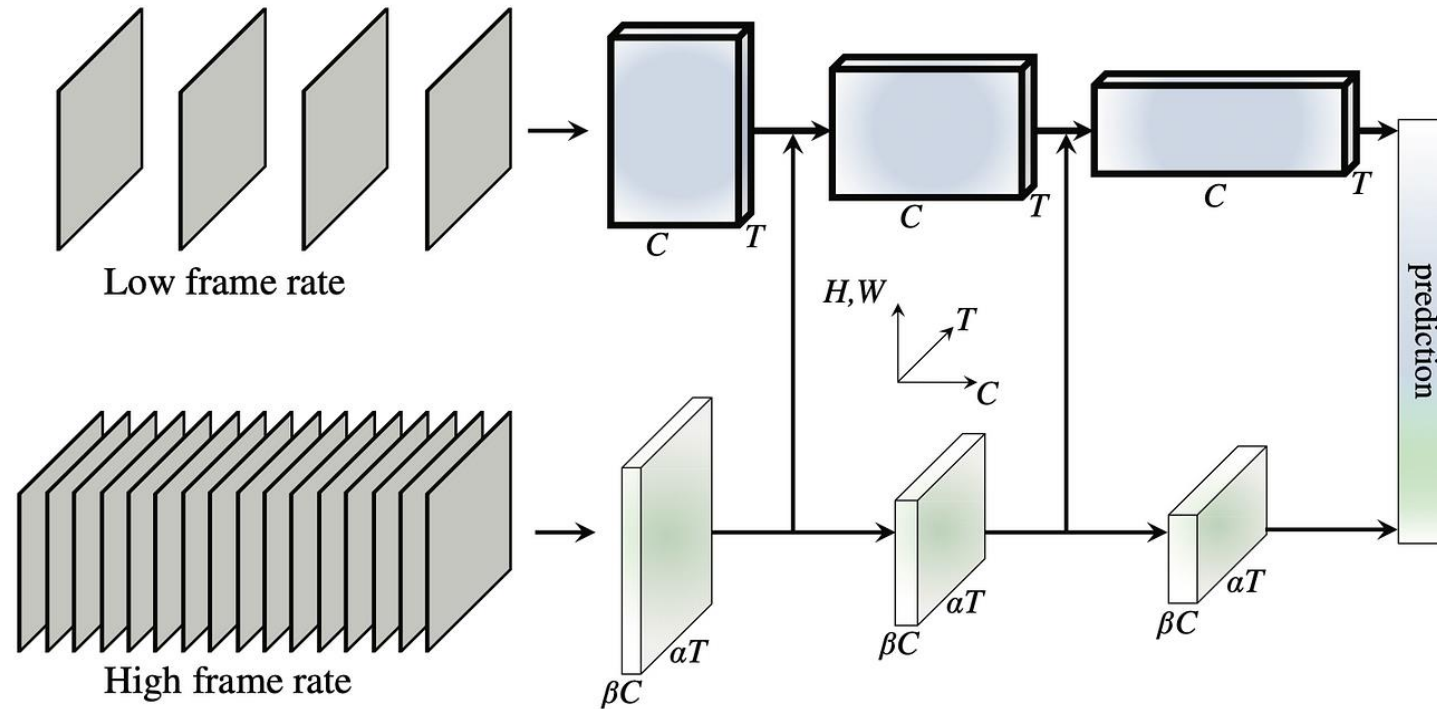
COMPARATION OF THE MODELS



LSTMS FOR VIDEO

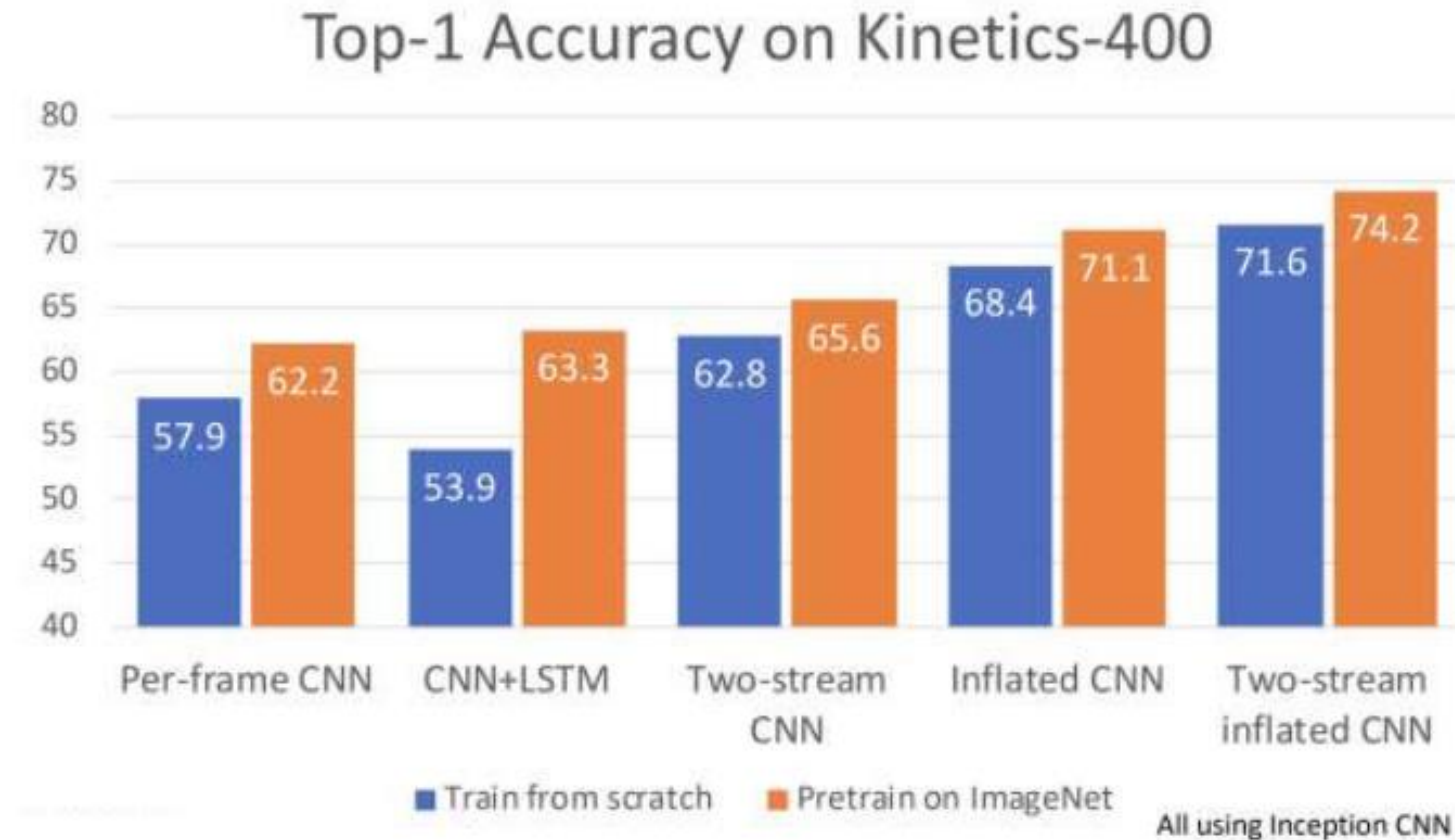


SLOWFAST NETWORKS



- Idea: Combine slow (spatial information) and fast (action information) temporal streams.
- The fast path processes α times more frames than the slow path. Typically $\alpha = 8$.
- The fast path has β times more channels than the slow path. Typically $\beta = 1/8$.

COMPARATION OF THE MODELS



OTHER PROBLEMS IN VIDEO

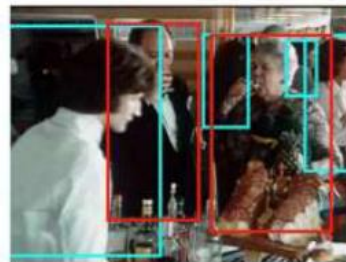
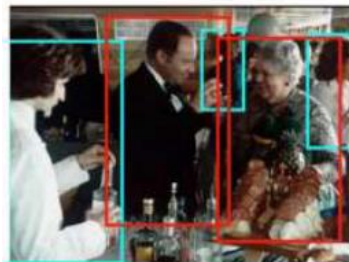
Temporal Action Localization (TAL)

Running

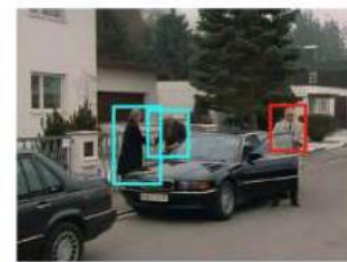
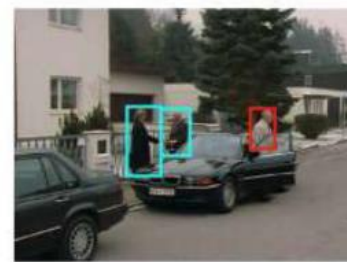
Jumping



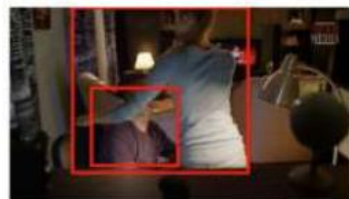
Spatio-Temporal Action Detection



clink glass → drink



open → close



grab (a person) → hug



look at phone → answer phone



T H A N K S !

